重力のイメージ

keyword: F=ma, free fall, air resistance,
Kepler, Law of Harmonies

source: kepler_mahmonic, SolarSystem.mw, d1_python_ode.ipynb

daddygongon 26/5/31

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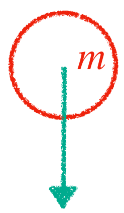
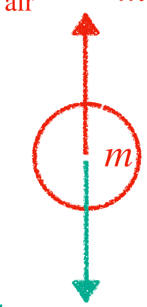
$$F = ma \qquad F_{\mathrm{air}} = -mCv$$





$$-mg = m\frac{dv}{dt} \qquad -mg - mCv = m\frac{dv}{dt}$$

$$\frac{v_{i+1} - v_i}{dt} = -g \qquad \frac{v_{i+1} - v_i}{dt} = -g - Cv$$

$$F_{\mathrm{Earth}} = -mg \qquad v_{i+1} = v_i - g\,dt \qquad v_{i+1} = v_i + (-Cv - g)\,dt \qquad F_{\mathrm{Earth}} = -mg$$

```
def euler_free_fall(x0, v0):
    v1 = v0 - g * dt
    x1 = x0 + v0 * dt
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def euler_with_air_resistance(x0, v0):
    v1 = v0 + (-cc * v0- g) * dt
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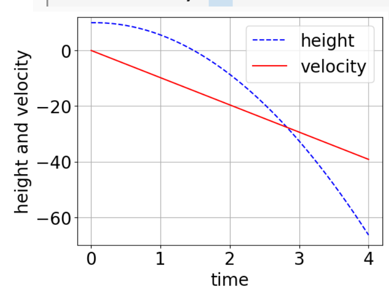

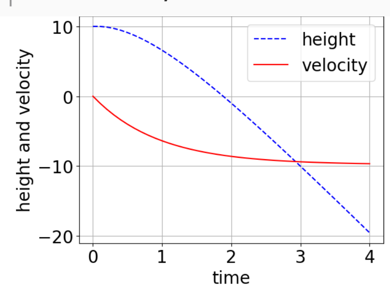

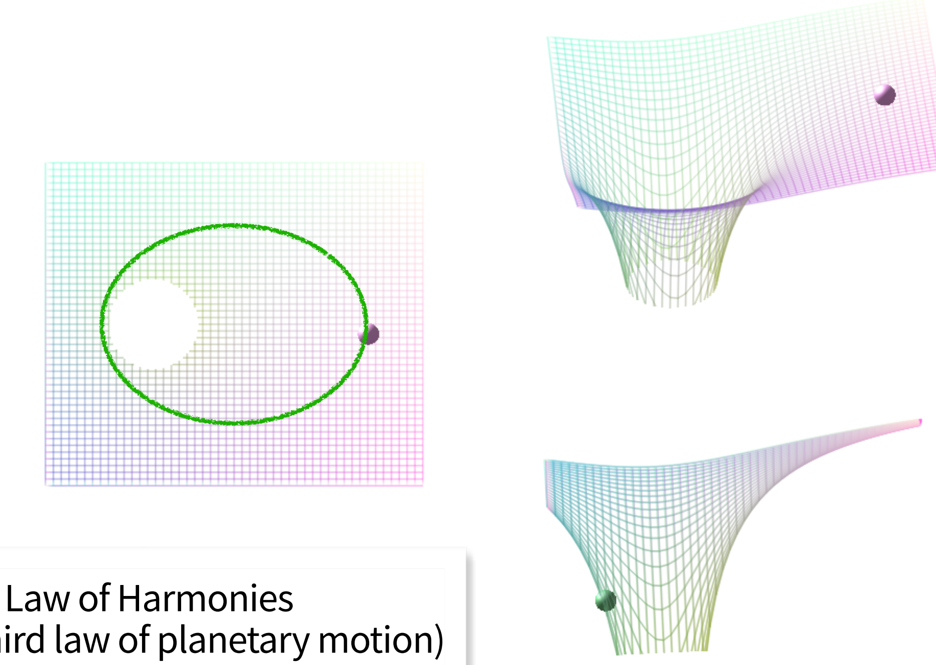

調和の法則 Law of Harmonies (Kepler's third law of planetary motion)

惑星	公転周期(T)	軌道長半径(r)	T^2/r^3 by Copilot	T^2/r^3 by you
水星	0.241	0.387	0.037	
金星	0.615	0.723	0.078	
地球	1.000	1.000	1.000	1.000
火星	1.881	1.524	0.296	
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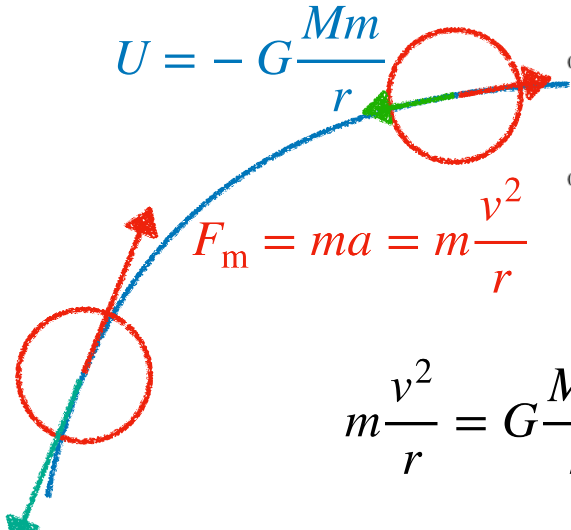
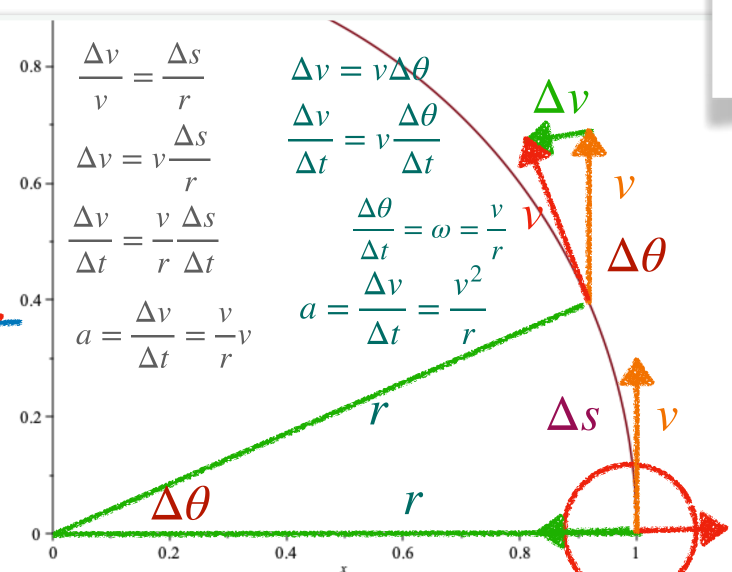


$$U = -G\frac{Mm}{r}$$

$$F_{\mathrm{m}} = ma = m\frac{v^2}{r}$$



$$\frac{\Delta v}{v} = \frac{\Delta s}{r} \qquad \Delta v = v\frac{\Delta s}{r} \qquad \frac{\Delta v}{\Delta t} = \frac{v}{r}\frac{\Delta s}{\Delta t} \qquad a = \frac{\Delta v}{\Delta t} = \frac{v}{r}v$$

$$\Delta v = v\Delta\theta \qquad \frac{\Delta v}{\Delta t} = v\frac{\Delta\theta}{\Delta t} \qquad \frac{\Delta\theta}{\Delta t} = \omega = \frac{v}{r} \qquad a = \frac{\Delta v}{\Delta t} = \frac{v^2}{r}$$

$$m\frac{v^2}{r} = G\frac{Mm}{r^2} \qquad v^2 = G\frac{M}{r}$$

$$F_{\mathrm{M}} = -\frac{dU}{dr} = -G\frac{Mm}{r^2} \qquad T = \frac{2\pi r}{v} \qquad T^2 = \frac{4\pi^2 r^2}{v^2} = \frac{4\pi^2 r^3}{GM}$$

$$\vec{r} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} r\cos(\omega t) \\ r\sin(\omega t) \end{pmatrix}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \begin{pmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{pmatrix} = \begin{pmatrix} -r\omega\sin(\omega t) \\ r\omega\cos(\omega t) \end{pmatrix}$$

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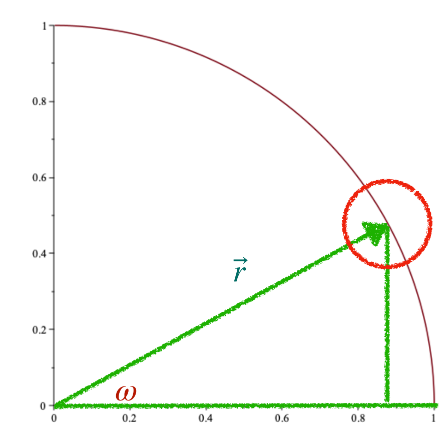
$$\vec{a} = -\omega^2\begin{pmatrix} r\cos(\omega t) \\ r\sin(\omega t) \end{pmatrix} = -\omega^2\vec{r}$$

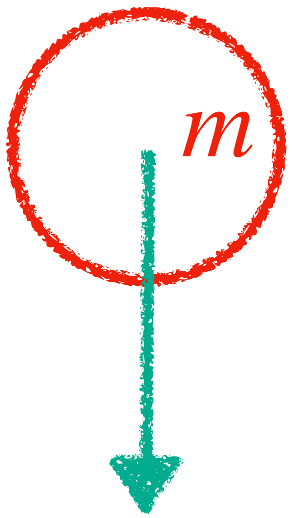


$$v = r\omega$$

$$a = r\left(\frac{v}{r}\right)^2 = \frac{v^2}{r}$$

$$F_{\text{air}} = -mCv$$

$$F = ma$$



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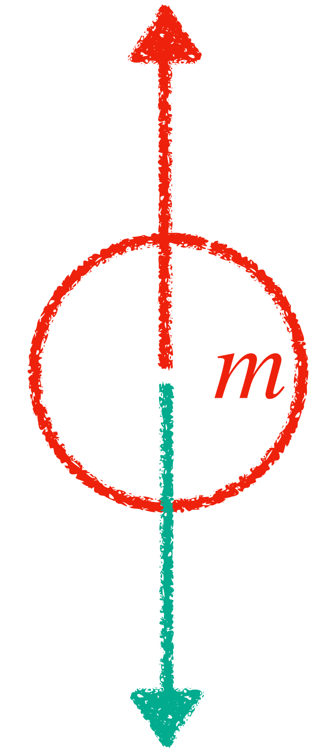

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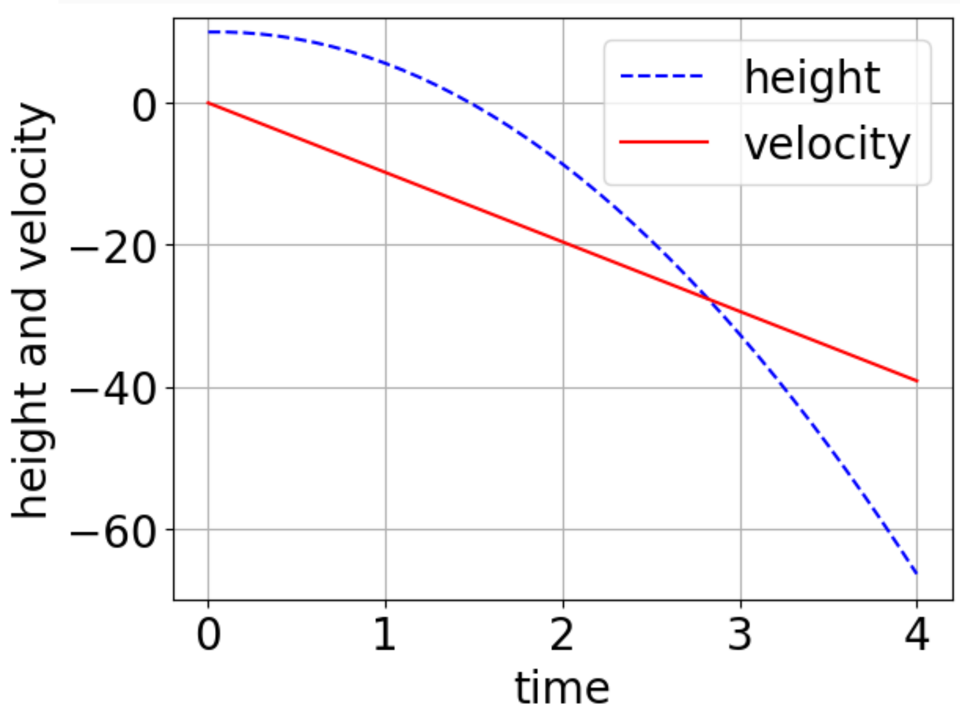

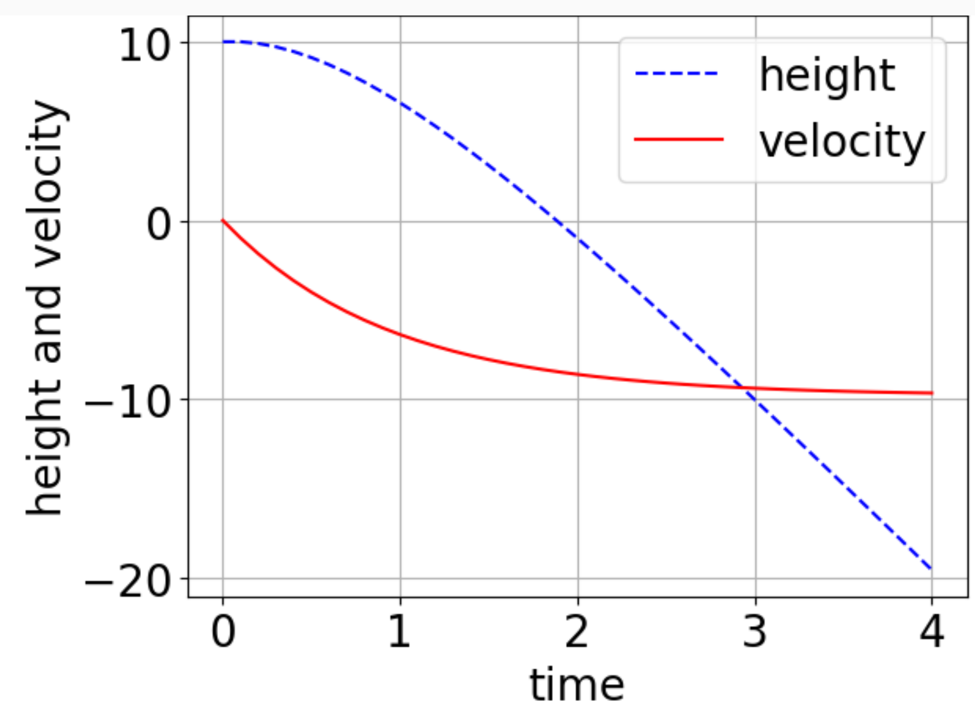

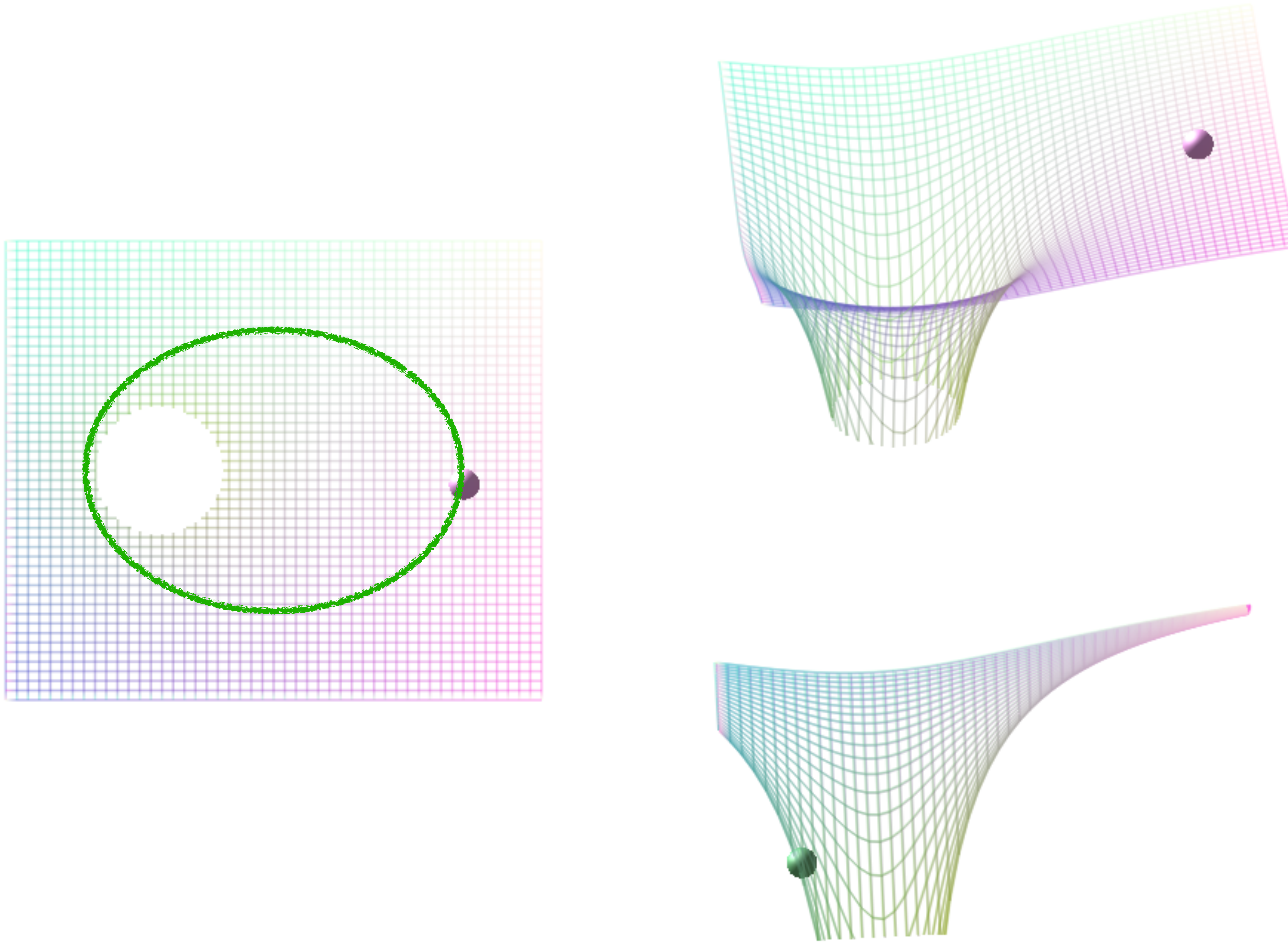

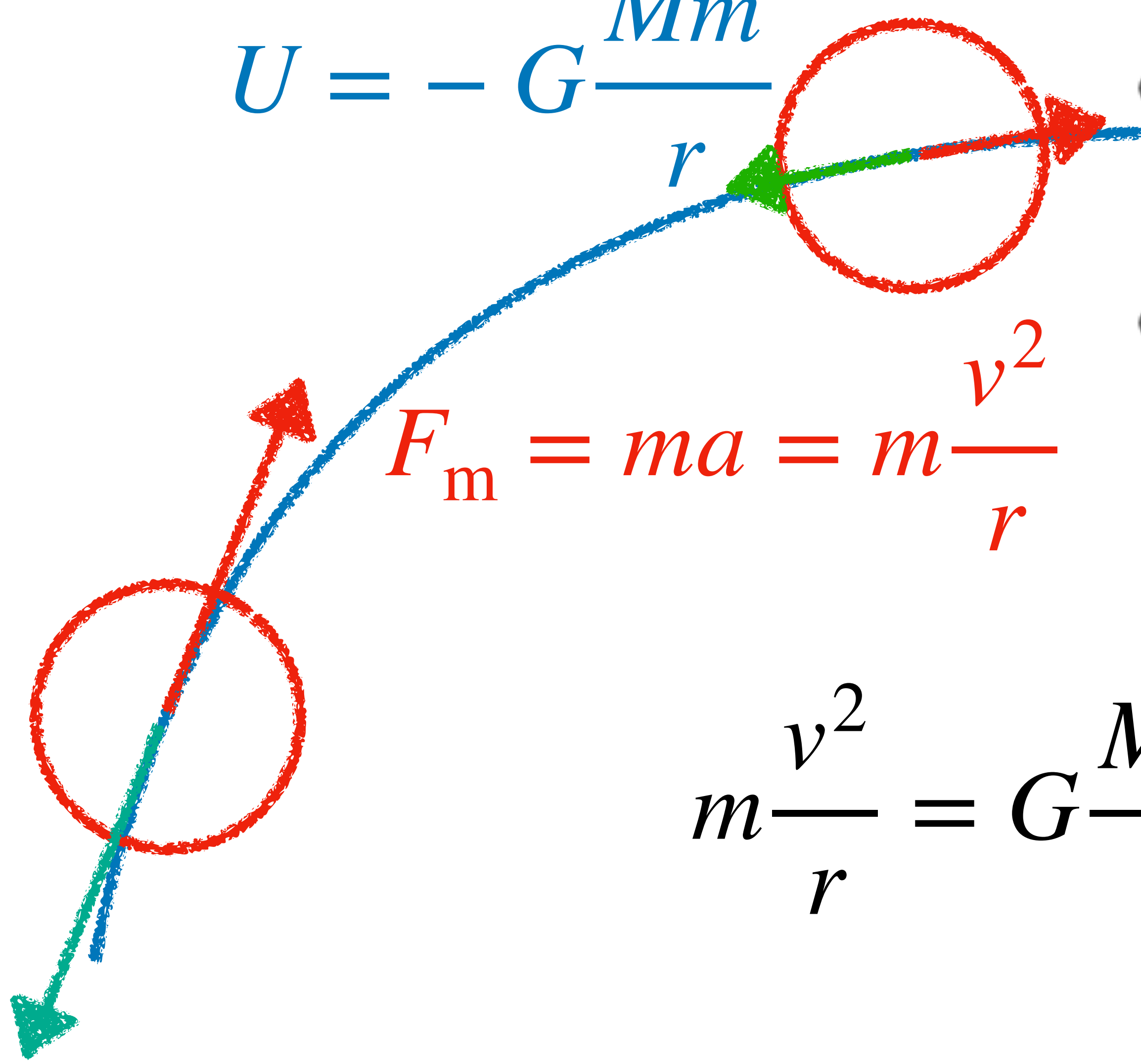

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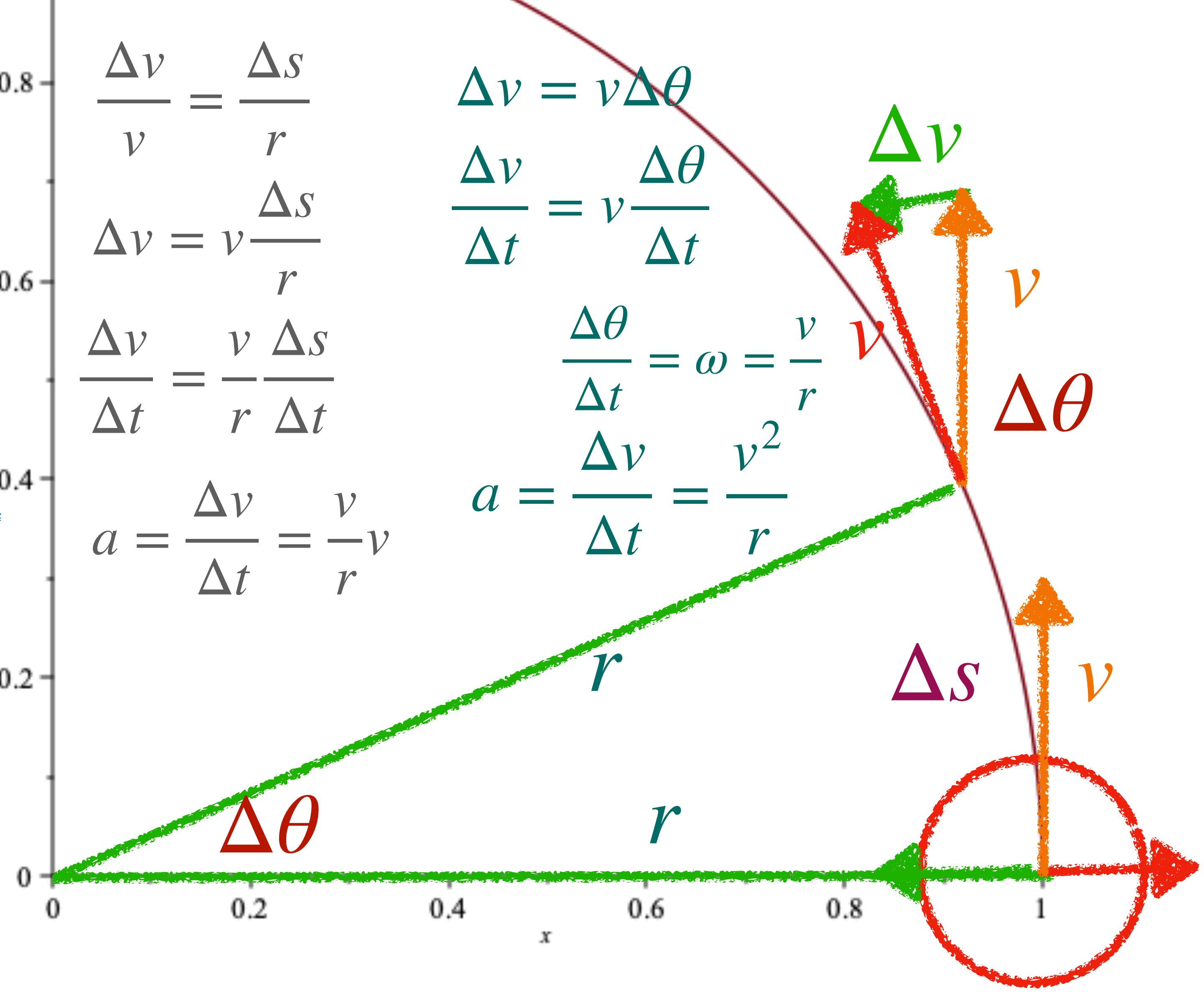

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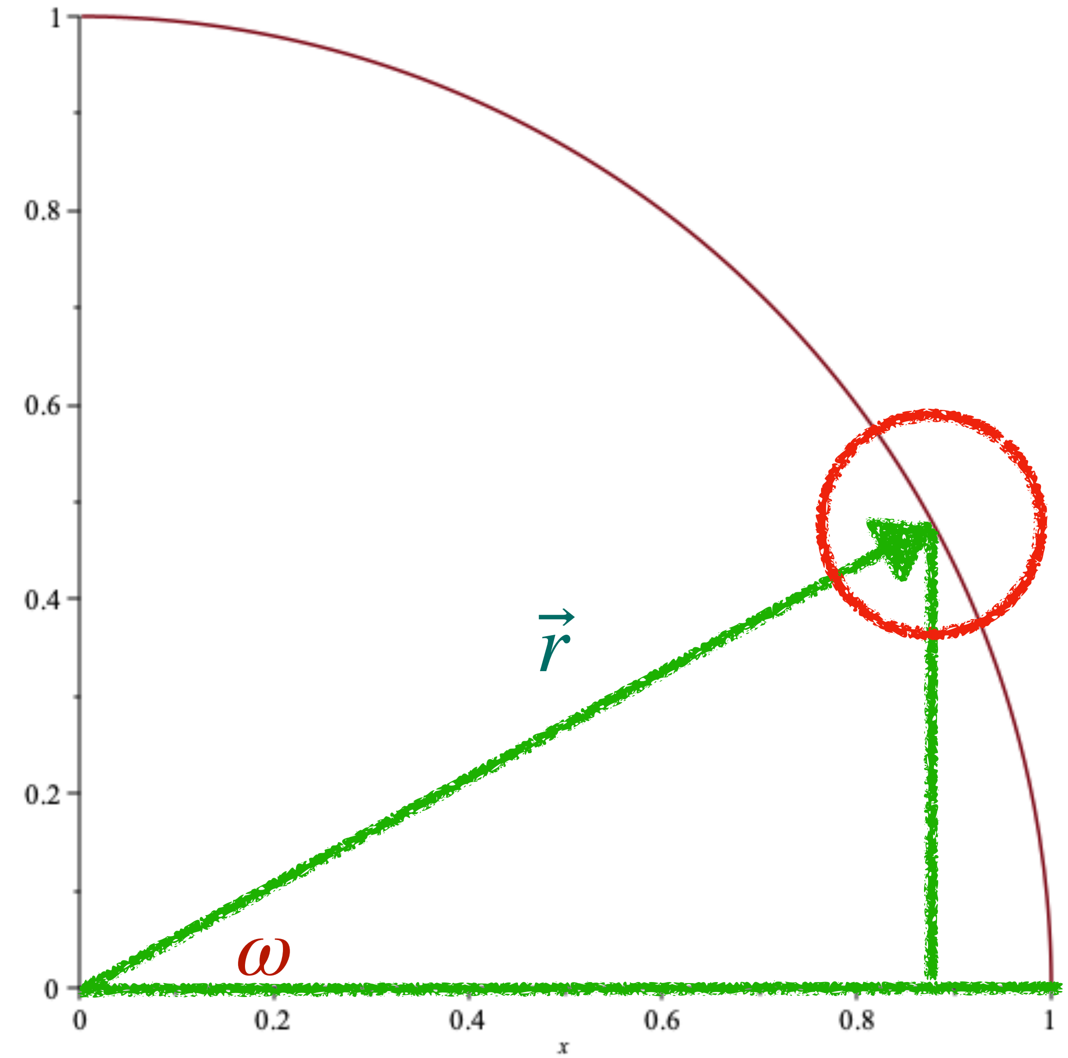

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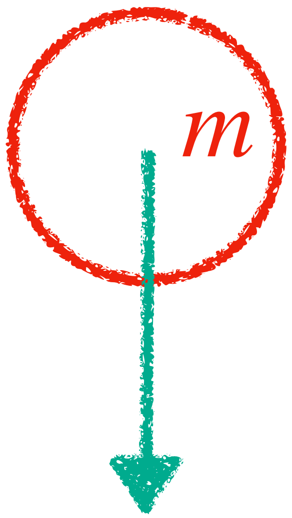

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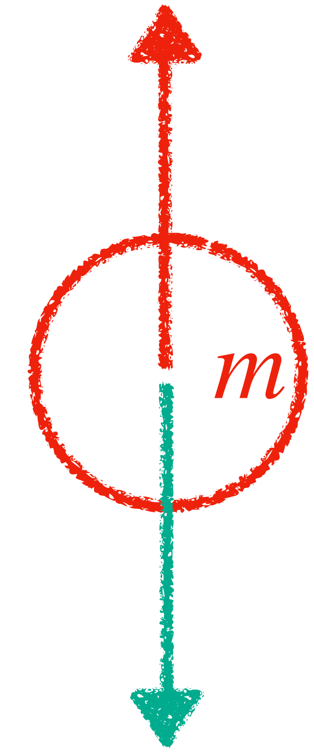

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